

Avatar-Driven Interaction: Constructing Immersive Education in Virtual Classrooms

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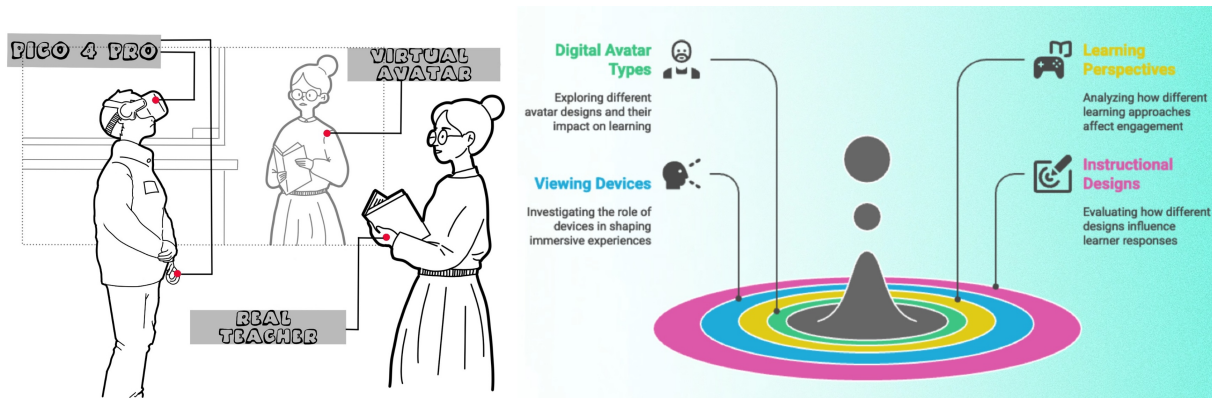


Figure 1: Virtual reality teaching setup illustrating the interaction among a student using VR technology, a virtual avatar instructor, and a real teacher in a blended learning environment.

ABSTRACT

With advances in multimedia and computer graphics technology, 3D virtual avatars have demonstrated promising performance in education. However, existing research faces challenges in systematically studying student interaction. In this study, a Virtual Classroom Interaction and Modal Assessment (VCIMA) is proposed to integrate various factors to explore the impact on learners' sense of presence, engagement, and preference. Specifically, we analyze key digital avatar types, learning perspectives, viewing devices, and their roles in shaping immersive learning experiences. Then, five elaborate scenarios around a physical demonstration are introduced to simulate diverse learning contexts, assess multimodal interactions, and evaluate how different instructional designs influence learner responses. Subsequently, a substantial corpus of experiments has been conducted under different conditions to verify the effectiveness of VCIMA and reveal that the adaptive multimodal feedback of 3D avatars in virtual reality enhances participation and presence. These findings provide actionable insights for designing future virtual learning environments, emphasizing the necessity of personalized interaction models.

Index Terms: virtual avatars, human-computer interaction, user studies, virtual reality

1 INTRODUCTION

Avatar technology has attracted increasing attention in educational research with the significant development in multimedia and computer graphics technology, and 3D virtual digital people have driven the rise of the virtual classroom. VR applications often support the embodiment of a user through virtual avatars, making avatars an

integral part of VR environments. We see the use of avatars in various contexts (such as gaming [40], training [14], and socialization [15]). The utilization of high-fidelity avatars has been demonstrated to enhance engagement and presence in virtual reality (VR) and augmented reality (AR), thereby prompting research into the potential of these technologies to improve conventional teaching methods [33].

The emergence of comparative research across various models, including but not limited to VR 2D avatars, VR 3D avatars, 2D avatars, and real teachers, further highlights the significant impact of these advances in education [6]. VR platforms such as 2D Engage VR [32] and 3D VictoryXR [27] facilitate the delivery of educational content within virtual environments, with the assistance of simulated instructors. Beyond the confines of academia, the pioneering work of virtual digital people and virtual classrooms provides an inspiring foundation for innovation in educational technology [2]. Virtual instructors on [29] and VRChat [38], for example, engage students via dynamic lessons powered by Unreal Engine 5. By integrating high-fidelity 3D avatars with immersive learning environments, these technologies increase student engagement and presence, offering new possibilities to deliver personalized, standardized instructional content [19]. For conceptual clarity, we defined the core variables as follows: (1) interactivity refers to the degree of control the user has over the avatar, ranging from passive observation to active manipulation via the controller; (2) perspective refers to the spatial viewpoint that mediates the presence of peers (first-person avatar vs. third-person observation); and (3) presentation format includes the mode of instruction delivery (immersive VR, 2D display). These dimensions were chosen based on their impact on cognitive load (Sweller, 2020) and social presence (Biocca et al., 2003). They form the basis of our controlled experimental study of the virtual classroom experience.

Virtual classrooms offer enhanced interactivity and flexible perspectives through VR headsets, standard screens, and hybrid platforms. With VR's growing adoption globally, understanding its impact on student engagement becomes crucial. While existing research examines VR's potential for enhanced immersion and mem-

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orization (De Freitas and Neumann, 2009), virtual classrooms combining digital teachers with immersive design represent a distinct category. This raises the critical question: Does virtual reality effectively improve student engagement, immersion, and learning outcomes? Comprehensive exploration of VR’s impact in virtual classroom environments is therefore essential.

This paper proposes the **Virtual Classroom Interaction and Multimodal Assessment (VCIMA)** framework to investigate how virtual human–computer interaction, peer perspectives, and pedagogical modalities influence student engagement, presence, learning outcomes, and learner preferences in virtual classrooms. Rather than introducing a new pedagogical theory or interaction technique, VCIMA is positioned as an analytical and comparative framework that organizes these design dimensions within a unified experimental structure. In contrast to prior studies that typically examine interactivity, spatial perspective, or presentation modality in isolation, this work experimentally examines their combined influence on learning experiences. Five simulated instructional scenarios compare immersive VR with traditional methods. A user study distills practical design insights for developing next-generation virtual learning environments. The main contributions are summarized as follows:

1. We proposed the **Virtual Classroom Interaction and Modal Assessment (VCIMA)**, which integrates interactivity, perspective, and modality variations to evaluate and optimize student experiences in virtual learning environments.
2. We introduced five elaborate scenarios around a physical demonstration to form a controlled experimental user study and distilled practical design insights to develop cutting-edge virtual learning ecosystems that redefine immersive, inclusive, and adaptable education on a global scale.
3. Extensive experiments and simulations under various conditions provide compelling evidence that interactive virtual classrooms with dynamic perspectives significantly enhance student engagement and presence. Based on these findings, design recommendations are proposed to improve immersive theatre experiences in virtual reality contexts.

2 RELATED WORK

2.1 Virtual Reality as a New Teaching Space

In today’s fast-evolving educational landscape, the demand for flexible, engaging learning environments has propelled virtual reality (VR) classrooms into the spotlight [13]. Moving beyond static video lectures, VR education now harnesses cutting-edge AR, VR, and MR technologies to craft immersive teaching spaces [34]. Platforms like Engage VR bring virtual instructors to life for interactive science labs, while VictoryXR immerses students in historical reenactments with stunning detail. Unlike traditional classrooms, VR offers unobstructed views, letting students shift angles say, zooming into a virtual engine or orbiting a 3D model and feel closer to the lesson [39]. Stanford’s VR lecture halls, for instance, place students “next” to a digital professor, enhancing connection [1]. However, VR is not flawless; researchers note it lacks the full sensory depth of physical spaces [25], necessitating rich visual, auditory, and haptic feedback for true immersion [5]. Comfort during extended VR sessions also matters, as Kim et al. emphasize, underscoring the need for practical, user-focused VR research [9].

Therefore, virtual teaching has various forms in terms of viewing perspectives. However, despite numerous and widespread cases, there are various forms of virtual teaching in terms of viewing perspectives. To fill the gap in the literature, in this study, we analyzed which one is more preferred by students, a single perspective or multiple open perspectives (including classmates raising their hands)

2.2 Teaching with Virtual Avatars

Integrating virtual avatars into education has long intrigued pioneers, with early experiments shaping today’s advancements. In the early 2000s, Stanford educators tested virtual tutors in simulated labs using basic VR headsets [12], while the University of Central Florida’s TLE TeachLivE introduced avatar students for teacher training [10]. Concurrently, Second Life [4] enabled virtual lectures for remote learners, and Zhejiang University in China piloted holographic instructors for engineering courses [37]. By 2018, Harvard and Zhejiang collaborated on RUM, using VR avatars to teach anthropology across continents, blending global classrooms with virtual embodiment [22]. These efforts laid a foundation for transforming education through virtual interfaces, culminating in immersive virtual avatar teaching systems, as illustrated in Fig. 1.

Recent multimedia and VR advancements have elevated virtual teaching’s scope and realism. VictoryXR offers VR biology lessons with lifelike avatar instructors on Oculus. Korea’s EBS Virtual Classroom uses motion capture for hybrid live-virtual lessons, viewable via PC or VR [31]. Beyond academia, Virtual Speech employs AI-driven avatars for medical role plays [3].

Rapid advances in VR, driven by platforms such as Unreal Engine 5, have introduced 3D virtual instructional avatars embodied in immersive classrooms, simulating classroom-based learning experiences. Unlike 2D video or 2D VR avatar content, 3D VR avatars enable spatial engagement with virtual instructors and peers. This study examines how avatar interactivity (responsive versus non-interactive) and interface design (singular versus open perspectives) affect engagement, presence, learning outcomes, and preferences. By comparing VR modalities with traditional 2D and real-world teaching, this research assesses the potential of interactive VR to enhance or replace conventional pedagogy and explores its strengths and limitations.

3 ANALYSIS OF VIRTUAL AVATAR TEACHING

3.1 Viewing Formats of Virtual Avatar Teaching

In comparison with the conventional classroom environment, students have the opportunity to experience virtual avatar teaching based on VR from a variety of perspectives. Students have the opportunity to experience VR-based avatars from different perspectives compared to traditional classroom environments. The experimental conditions were designed using a review of current viewing methods from two perspectives: Viewing device and Visual angle.

Viewing device. Depending on educational resource limitations, virtual avatars teaching can be experienced through various devices, roughly categorized into traditional 2D displays and immersive VR hardware [36]. (1) *Traditional 2D displays* are widely used in situations where VR integration is limited or impractical. (2) *VR devices* are predominantly employed in specialized educational environments designed to maximize immersion, thereby offering students a more dynamic and spatially rich classroom experience. While VR adoption grows rapidly in education, 2D displays become a more common default due to limitations in access and technical barriers. This disparity underscores delivery-perception gaps in avatar teaching, driving the need to examine viewing perspectives.

Visual angle. This study sets out to explore and contrast three distinct VR teaching perspectives. (1) *Round-table approach*: Collaborative immersive seating with peer gesture interaction enhances social presence through optimized group visibility and dynamics, effectively fostering learning communities. As shown in Fig. 2a. (2) *Row-column approach*: Linear spatial arrangement balances focused attention and social cue perception, leveraging environmental familiarity to maintain structured engagement. As shown in Fig. 2b. (3) *Fixed approach*: Content-centric minimalism maximizes cognitive clarity through sensory reduction, strategically supporting the passive absorption of lecture materials, as shown in Fig. 2c. The discrepancy between different visual perspectives emphasizes



(a) Round-table approach



(b) Row-column approach



(c) Fixed approach

Figure 2: Comparison of different VR viewing perspectives in virtual classroom environments

the importance of purpose-driven perspective design in the virtual classroom. Therefore, it is essential to experimentally investigate the effects of VR viewing perspectives on immersion, engagement, and learning preferences.

3.2 Avatar types

This study also focuses on the effect of the type of avatar in a virtual reality environment on the learning experience. The independent variable "avatar type" includes two cases: 3D interactive avatars and 2D non-interactive avatars. (1) *3D interactive avatars* employ Natural Language Processing (NLP) and Text To Speech (TTS) [30] technologies to enable the avatar to generate speech feedback. Students interact via voice or text during Q&A sessions, with voice commands triggering real-time avatar actions and responses. They can modify commands at any time for updated feedback [21]. Prior research through VR experiments validated its immersive benefits. As shown in Fig. 3. (2) *2D non-interactive avatars* hinder real-time student engagement and learning due to their unidirectional instructional approach, as shown in Fig. 4. Classroom interactivity suffers from the avatar's lack of adaptability to student feedback, potentially resulting in distraction and decreased motivation. The variation among virtual avatar types underscores the necessity for multimodal interactive design in educational settings. Hence, an experimental investigation into the immersion provided by VR avatars, alongside students' engagement and learning preferences, is imperative.

However, they may face challenges in emotional connection and complex feedback, including responding to learners' emotional needs or providing personalized instruction [20].



Figure 3: 3D avatar

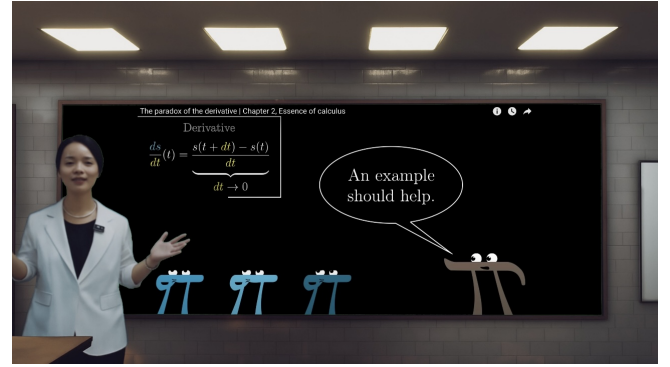


Figure 4: 2D avatar

4 EXPERIMENTAL DESIGN

4.1 Study Design

We identify two main experimental factors to address the research questions: (1) *viewing formats*: viewing medium and viewing perspective. (2) *avatar types*: 3D interactive avatars and 2D non-interactive avatars. The row-based and round-table layouts used in our virtual classroom scenarios are derived from common physical classroom arrangements in higher education. Rather than re-evaluating student reactions in physical classrooms, this study adopts these established layouts as reference baselines and focuses on how their virtual implementations interact with avatar embodiment and presentation modality. The combination of these factors resulted in various experimental conditions, including different levels of interactivity, variations in the viewing perspective in VR, and different presentation modes in all groups. We conducted a user study using a mixed-subject approach with two viewing styles and two avatar types as within-subject variables and two viewing devices as between-subject variables.

4.2 Implementation of Experimental Environment

The experimental environment, developed using Unreal Engine, comprised four instructional conditions: 3D Avatar Condition featuring immersive VR with voice response and gesture recognition across three viewpoint configurations (fixed, round-table, grid layouts); 2D Avatar Condition using pre-recorded video content within the same VR interface; Mobile 2D condition deploying simplified video on tablet devices; and Real Classroom Condition employing traditional instructional equipment. The VR conditions utilized PICO 4 PRO headsets with NVIDIA RTX 3070 GPU for real-time rendering, integrated PICO SDK for interactive functionality with

real-time engagement tracking, and underwent stability verification ensuring sub-20ms latency. Pre-experiment familiarization sessions and counterbalanced condition sequencing were implemented to control for order effects. Among these instructional settings, the present study focuses on a subset of five core experimental scenarios for quantitative analysis, selected to enable controlled comparison of avatar embodiment and classroom layout within virtual learning environments. Other instructional conditions, including the Real Classroom and Mobile 2D settings, are included solely as supplementary baseline references to support qualitative interpretation and contextual comparison, rather than as primary factors in the statistical analysis.

4.3 Participants

One hundred fifty university students from different classes participated in this study, using a 5-group experimental design with 30 students per group. Participants had a mean age of 21.7 years ($SD = 1.8$), with balanced gender distribution (75 females, 75 males), and 90 participants reported previous virtual reality experience. The experiment employed a mixed factorial design, where equipment type served as a between-subjects factor, while avatar type and view-point layout served as within-subjects factors. Participants were assigned to experimental conditions through random grouping to ensure balanced baseline characteristics across groups. All groups were instructed by the same educator with over twenty years of teaching experience. The study received approval from the Academic Ethics Committee of the affiliated institution. Additionally, participants were informed of their right to withdraw from the study at any time without any impact on their rights.

4.4 Procedure

Participants engaged in various classroom scenarios involving avatars or real teachers through 2D displays, VR devices, or live interactions. A 24-inch Dell monitor was utilized for traditional 2D displays, while a PICO 4 PRO VR headset was employed for VR conditions. The Real Teacher condition occurred within a controlled classroom setting. All virtual scenes in 2D and VR conditions were confirmed to operate at 72 fps or higher for optimal performance. The presentation style was adjusted according to participants' prior VR experiences. Additionally, the sequence in which participants experienced the scenes was randomized to mitigate learning effects or familiarity bias.

This study experimentally evaluates the impact of different collaborative environments on educational outcomes through five core experimental scenarios (including 3D/2D avatars, round table/row and column layouts, and fixed-space designs) while controlling for equipment, spatial layout, and avatar presentation, as shown in Figure 6. The specific scenarios are as follows:

(a) **VR-Roundtable layout-3D avatars:** Users sit around a virtual roundtable with 3D avatars in a VR environment to simulate a natural face-to-face conversation scene and evaluate the effect of roundtable collaboration.

(b) **VR-Row layout-3D avatars:** 3D avatars are arranged in classroom-style rows and columns to test the effect of a structured spatial layout on collaboration efficiency.

(c) **VR-Fixed layout-3D avatars:** 3D avatars are fixed in position to investigate the effect of reducing the need for spatial movement on user experience.

(d) **VR-Fixed layout-2D avatars:** 2D avatars is used to evaluate the impact of avatars presentation on immersion.

(e) **2D Screen-Fixed Layout-2D avatars:** A traditional 2D interface presenting a fixed layout is used as a baseline control group for the VR experiment

Educational content. The instructional session focused on introductory calculus, specifically derivative-based limit evaluation

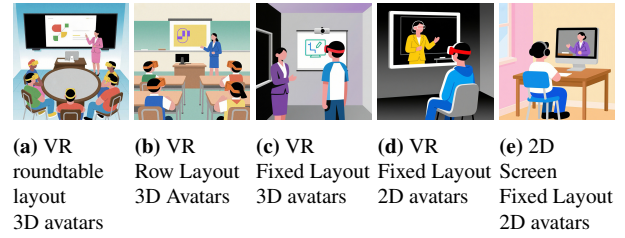


Figure 5: Overview of the five experimental scenarios.

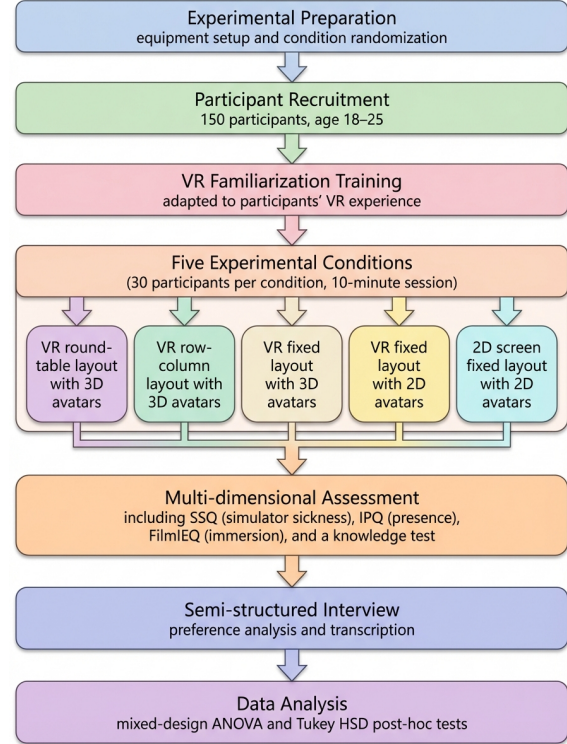


Figure 6: Experimental procedure.

using L'Hôpital's rule. The lesson emphasized procedural reasoning, including identifying indeterminate forms, applying derivative rules step by step, and interpreting results. Representative problems included limits involving exponential, logarithmic, and trigonometric functions, consistent with standard undergraduate calculus curricula. The instructional content was aligned with the subsequent knowledge test, which assessed both factual recall of derivative rules and the correct procedural application of L'Hôpital's rule in limit problems.

This study employed a standardized 10-minute VR instructional session with varied avatar conditions, followed by questionnaires (SSQ/IPQ/FilmIEQ) and a knowledge test to assess discomfort, presence, immersion, and learning preferences. All questionnaire items were rated on a 7-point Likert scale. Specifically, simulator sickness was measured using the Simulator Sickness Questionnaire (SSQ) [8], covering nausea, oculomotor, and disorientation symptoms (e.g., "nausea", "eyestrain", "dizziness"). Presence was measured using the Igroup Presence Questionnaire (IPQ) [26], including general presence, spatial presence, involvement, and experienced realism (e.g., "I had a sense of being there in the virtual classroom", "I felt involved in the virtual lesson"). Immersion was measured using the FilmIEQ [23], a media-adapted

version of the Immersive Experiences Questionnaire (IEQ) [24], focusing on attentional engagement and dissociation from the real world (e.g., “I lost track of time while participating”, “I felt detached from my physical surroundings”). The knowledge assessment comprised 12 closed-book questions, including 10 multiple-choice items targeting derivative rules and limit conditions, and two application problems requiring step-by-step use of L’Hôpital’s rule. Finally, semi-structured interviews were conducted to collect participants’ preferences across five instructional environments, with audio transcriptions analyzed. The complete experimental procedure was completed within approximately 40 minutes.

5 RESULTS

5.1 Quantitative analysis

We analyzed the survey responses using a two-way mixed ANOVA. This mixed-design ANOVA treated the presentation mode (VR, 2D display) as a between-subject factor. In contrast, the type of avatar (3D vs. 2D avatar) and viewpoint (fixed, round table, and row-column) were treated as within-subject factors. Additionally, when there were significant interaction effects between factors, we conducted a post hoc analysis using Tukey’s HSD to confirm significant differences between groups.

5.1.1 Simulator Sickness (SSQ)

The device type significantly affected simulator sickness (SSQ) scores ($F(1, 148) = 18.42, p < 0.001$). Participants reported higher sickness levels in VR ($M = 3.88$) than in 2D ($M = 2.5, p < 0.001$). Additionally, avatar type significantly impacted SSQ scores ($F(1, 148) = 7.15, p = 0.010$). Participants exhibited lower SSQ scores with a 3D avatar ($M = 3.9$) versus a 2D avatar ($M = 3.15, p = 0.010$). Moreover, layout configuration significantly affected simulator sickness ($F(2, 148) = 5.67, p = 0.006$). The RoundTable layout ($M = 3.1$) resulted in lower SSQ scores than the RowColumn layout ($M = 4.6, p = 0.005$), indicating the influence of spatial arrangement on VR discomfort. Avatar dimensionality significantly influences the alleviation of simulator sickness in virtual environments. In contrast, the perspective type (fixed, round-table, or row-column) and its interaction with presentation mode do not notably impact sickness levels. As shown in Fig. 7a. Participants who scored higher on the SSQ scale completed the entire experiment successfully despite experiencing discomfort in the middle of the experiment. It is worth noting that SSQ scores across all conditions remained relatively low (all mean values below 6), indicating mild or negligible simulator sickness symptoms rather than clinically significant discomfort. Accordingly, SSQ is interpreted in this study as a comparative indicator of relative comfort across conditions, rather than as evidence of severe sickness. The low SSQ values also suggest that the experimental setup and session duration were generally well tolerated by participants.

5.1.2 Sense of Presence

An analysis of the overall score and the sense of presence subscale was conducted to assess the sense of presence. As shown in Fig. 7b. It was found that viewing the medium affected the sense of presence. However, the study revealed no significant main effects for all factors related to social realism. As shown in Table 1 and Table 2

The effect of the viewing medium: The viewing medium significantly impacted presence scores ($F(1, 148) = 15.29, p < 0.001$). Post-hoc analyses indicated that VR conditions ($M = 5.03, SD = 0.96$) yielded higher IPQ scores than 2D conditions ($M = 4.39, SD = 0.60; p < 0.001$). The consistent observation across experimental conditions indicates that VR technology significantly augments the sense of presence relative to 2D displays, suggesting

that the medium of viewing critically affects presence score variability, supporting the hypothesis that VR’s immersive characteristics provide a markedly enhanced sense of presence compared to conventional 2D interfaces.

effect of the viewing perspective: Analysis indicated that layout type significantly affected IPQ scores ($F(2, 298) = 28.97, p < 0.0001$). Tukey HSD post-hoc tests revealed that the round-table layout ($M = 6.41$) surpassed all other layouts (all $p < 0.001$), indicating greater fidelity; fixed 3D avatar layouts ($M = 5.45$) were superior to the row layout ($M = 4.36, p < 0.001$) and fixed 2D avatar layout ($M = 3.73, p < 0.001$), suggesting enhanced immersion with 3D; the 2D screen condition ($M = 3.73, p < 0.001$) was significantly inferior to the round table layout ($p < 0.001$) yet superior to the fixed 2D avatar layout ($p < 0.05$), with no significant difference between row conditions ($p = 0.98$), indicating minimal effect of display method (VR vs. 2D screen) on IPQ in the row layout.

The effect of the type of avatar: The examination revealed that avatar classification exerted a substantial influence on presence awareness (IPQ) ($F(1, 149) = 12.63, p = 0.001$). Three-dimensional avatars ($M = 5.41$) markedly surpassed two-dimensional avatars across all evaluative conditions, with the divergence being most pronounced in the circular arrangement ($p < 0.001$) and the divergence being second most pronounced in the row-column arrangement ($p = 0.028$), while the fixed perspective condition also exhibited a significant advantage ($p < 0.001$). This discovery corroborates the considerable impact of 3D avatars on improving presence perception in interactive environments through enhanced spatial and depth cues.

Interaction between factors: The study demonstrated notable interaction effects among device type, layout configuration, and avatar dimensionality on IPQ scores ($F(2, 296) = 6.17, p = 0.004$). VR technology significantly elevated presence in roundtable setups ($M = 6.41$) compared to 2D displays ($M = 4.39, p < 0.001$), with the combination of VR, round-table layout, and 3D avatars yielding maximal IPQ scores, underscoring the importance of spatial configuration and avatar characteristics in enhancing user experience.

5.1.3 Sense of Immersion

Visual presentation medium had a significant effect on the effectiveness of environmental immersion, with ANOVA revealing substantial between-group differences. Post-hoc power analysis confirmed high sensitivity to detect differences in immersion across technological conditions. As shown in Fig. 7c.

The effect of the viewing medium: The findings indicated a notable influence of device type on reality dissociation (FilmIEQ) ($F(1, 148) = 28.36, p < 0.001$). The VR device ($M = 5.82$) significantly surpassed the 2D screen ($M = 2.45, p < 0.001$) in all configurations, with the round-table layout exhibiting the most pronounced VR advantage, while the fixed layout’s reduced VR advantage suggests that interactivity significantly amplifies VR’s immersive qualities, whereas 2D screens exhibited consistent but inferior performance due to the fundamental constraints of flat displays in promoting dissociation, potentially stemming from the enhanced spatial presence facilitated by stereo vision, head tracking, and spatial audio features unique to VR.

The effect of the viewing perspective: Analysis indicated layout configuration significantly affects real-world dissociation (FilmIEQ: $F(2, 298) = 15.83, p < 0.001$). The interactive round table configuration exhibited markedly superior dissociation scores ($M = 6.80$) relative to both row-column ($M = 3.80, p < 0.001$) and fixed ($M = 5.50, p = 0.003$) layouts, with the fixed layout also surpassing the row-column arrangement ($p = 0.012$). This trend was notably accentuated in the VR context, where the round-table configuration demonstrated a 58% enhancement over 2D screens, far

Table 1: Interaction Analysis Results for Viewing Media and Viewing Angle

Group A	Group B	Mean Diff	95% CI Upper	p-value	Cohen's d
RoundTable (VR)	RowColumn (VR)	2.05	2.87	<0.001***	1.68
RoundTable (VR)	Fixed (2D Screen)	2.68	3.45	<0.001***	2.15
Fixed (VR)	RowColumn (VR)	1.09	1.77	0.003**	0.95
RoundTable (VR)	Fixed (VR)	0.96	1.64	0.009**	0.83

***p < 0.001, **p < 0.01, *p < 0.05.

Table 2: Interaction Analysis Results for Avatars, Media and Viewing Angle

Group A	Group B	Mean Diff (IPQ)	95% CI Upper	p-value	Cohen's d
VR-3D-RoundTable	VR-3D-Fixed	0.96	1.64	0.009**	0.83
VR-3D-RoundTable	VR-3D-RowColumn	2.05	2.87	<0.001***	1.68
VR-3D-RoundTable	2D-2D-Fixed	2.68	3.45	<0.001***	2.15
VR-3D-Fixed	2D-2D-Fixed	1.72	2.49	<0.001***	1.48
VR-3D-RowColumn	2D-2D-Fixed	0.63	1.40	0.041*	0.55

***p < 0.001, **p < 0.01, *p < 0.05.

exceeding the fixed layout's 32% improvement, while 2D displays exhibited no significant layout-related variations (all $p > 0.05$). The substantial effect size implies that spatial interactivity is vital for promoting dissociation, likely via improved environmental encapsulation and social presence facilitated by 360° viewing and dynamic perspective shifts in the round-table setting.

The effect of the type of avatar: Data analysis revealed a significant effect of avatar type on environmental immersion (IEQ; $F(1, 149) = 15.72$, $p < 0.001$), with 3D avatars ($M = 5.45$) outperforming 2D avatars ($M = 3.73$, $p < 0.001$). This indicates 3D rendering enhances immersion, though no IEQ difference emerged in 2D screen conditions ($p = 0.18$), suggesting 3D avatars rely on stereoscopic depth cues in VR. Notably, 3D avatars showed greater IEQ gains in round-table vs. fixed layouts, implying spatial interaction amplifies their immersive effects. Results confirm avatar dimensionality's impact is context-dependent, requiring integration with display technology and spatial design for optimal immersion.

A significant interaction was observed among device type, viewing angle mode, and avatar regarding reality dissociation ($F(2, 296) = 9.27$, $p < 0.001$). VR devices enhanced dissociation scores by 37.5% in interactive discussions versus non-interactive formats ($p < 0.001$), while no significant difference was noted for 2D screens ($p = 0.18$). The VR advantages were maximized in interactive conditions ($p < 0.001$) but substantially reduced in fixed-viewing angle contexts ($p = 0.09$). 3D avatars augmented dissociation by 42.3% in VR ($p < 0.001$), but showed no significant impact on 2D screens ($p = 0.31$). The findings indicate that VR's spatial interactivity significantly amplifies dissociation experiences in dynamic scenarios, while 2D formats consistently exhibit minimal dissociation due to inadequate depth perception. Importantly, dissociation scores entered the highly immersive range only under the combined VR, round table, and 3D avatar conditions.

5.1.4 Learning Performance

This study explores the impact of experimental conditions on a 100-point knowledge assessment (10 MCQs, 2 computational problems). Findings show VR ($M=71.3$) outperforms 2D ($M=63.8$, $p<0.001$) by 11.76%, round-table layout ($M=81.3$) surpasses row-column/fixed layouts ($p<0.012$), and 3D avatars ($M=72.7$) excel over 2D ($p=0.004$). The optimal VR+round-table+3D avatar combination ($M=81.3$) improves scores by 27.4% versus baseline 2D ($p<0.001$). Computational problem-solving shows more pro-

nounced condition effects, indicating immersive environments enhance complex cognitive skills. The statistical results of the test scores are shown in Table 3.

5.2 Qualitative analysis

This section presents the results of the analysis of the post-experimental interviews. Opinions and preferences about the experimental factors and the reasons for these results are discussed. Participants' responses were categorised and summarised based on their perspective on viewing, the device type utilised, and the presence or absence of digital human interaction. In addition to the five core experimental scenarios, participants were also exposed to supplementary instructional settings (e.g., Mobile 2D and Real Classroom) for comparative discussion. These conditions are reported qualitatively to provide contextual insights into learner preferences, but are not included in the primary quantitative factor analysis.

5.2.1 Preference for Viewing Perspectives

The rounded table perspective performs better in VR 3D environments than fixed and rowcolumn perspectives. While initially aimed at promoting collaboration, the row and column perspective performed similarly to the fixed perspective, indicating a need for spatial layout optimization relative to task type in VR 3D. The unanimous preference for the round table perspective underscores its viability as a prominent configuration in future educational technology. The summary of experimenters' opinions on viewing perspectives in VR is shown in Table 4.

5.2.2 Preference for the Types of Virtual Avatar

This study shows participants prefer 3D virtual avatars, especially in VR round-table layouts, though real teacher scenes still compete in engagement. Layout, equipment, and avatar type significantly influence preferences, with immersive VR round tables and 3D avatars favored most. 3D avatars outperform 2D ones in VR, as the latter's simplicity can't match their effectiveness. Results call for optimizing VR avatar design to enhance user experience and indicate 3D avatars as the leading configuration for future edtech. The summary of experimenters' opinions on types of Virtual Avatar in VR is shown in Table 5

Table 3: Learning Performance Across Experimental Groups

Group	Layout and Avatar	Mean Total Score	Multiple-Choice (40)	Numeric (60)	Std. Dev.
G1	VR-Roundtable-3D	81.3	35.2	46.1	6.8
G2	VR-Row-3D	67.1	32.4	34.7	8.2
G3	VR-Fixed-3D	69.8	33.1	36.7	7.5
G4	VR-Fixed-2D	66.9	31.8	35.1	8.8
G5	2D Screen-Fixed-2D	63.8	30.5	33.3	9.1

***p < 0.001, **p < 0.01, *p < 0.05.

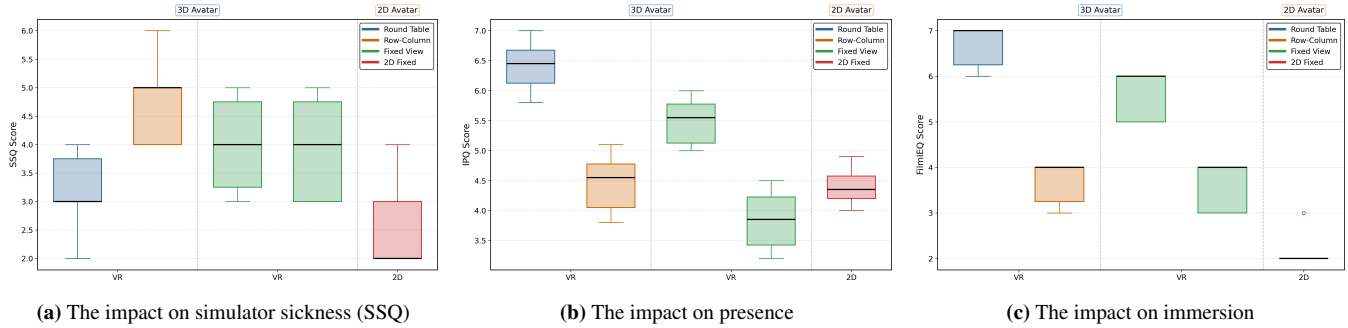


Figure 7: The results of analysis on viewing experience.

Table 4: Opinions on Roundtable, Fixed and Row-Column

Perspective	Advantages	Disadvantages
Roundtable	Promotes collaborative interaction and spatial immersion; encourages group discussions with multi-angle peer observation; the circular layout fosters inclusivity and peripheral awareness.	Visual obstructions due to seating; technical challenges in tracking movements; lack of a central focal point.
Fixed	Simple and easy to navigate; offers a stable viewpoint that supports sustained attention on the instructor; effective in 2D classroom-like settings.	Limited interactivity and sense of detachment; lack of environmental exploration; reliance on pre-rendered imagery.
Row-Column	Structured layout suitable for large audiences; ensures clear visibility of the central presenter.	Socially isolating and potentially disorienting; difficulty maintaining eye contact; technical issues such as avatar misalignment.

Table 5: Opinions on 3D and 2D Avatars

Aspect	Advantages	Disadvantages
3D Avatars	Widely perceived as immersive and naturally interactive; enhances peer communication; compatible with dynamic and spatial learning environments.	Inadequate rendering of fine avatar details; occasional stuttering during action execution.
2D Avatars	Intuitive design with minimal technical requirements; resembles conventional video conferencing tools for rapid adaptation; performs well on lower-spec devices.	Lack of depth reduces engagement and sense of presence; static expressions and audio synchronization issues; absence of immersive cues limits emotional involvement.

Table 6: Opinions on VR and 2D Display

Aspect	Advantages	Disadvantages
VR Devices	Creates immersive, realistic environments with natural interactions; enhances engagement via gesture/head tracking and tactile feedback; enables 360° spatial perception.	Causes motion sickness and neck fatigue; tracking errors and lack of physical references hinder multitasking; floating interfaces reduce the efficacy of question-solution dialogues.
2D Display	Offers easy operation, low cognitive load, and high multitasking efficiency; better eye comfort for long-term use.	Limited spatial expression and lack of true 3D interaction; weaker sense of presence reduces collaborative immersion; interaction is limited to basic gestures (click/drag).

Table 7: Design guidelines for virtual avatar teaching according to the content types.

Avatar Type	Roundtable View	Row-Column View	Fixed View
3D Avatar	Recommended	Not recommended unless: - Dynamic spatial anchor system (eye-tracking cognitive heatmaps) - Hovering identifiers in focus areas	Recommended with: - Dynamic background elements (e.g., floating curtains) - Perspective switching sync
2D Avatar	Conditionally recommended: - Trigger partial 3D rendering on key gestures (e.g., hand raise)	Not recommended unless: - Teacher demonstrates strong emotional execution	Conditionally recommended: - Speech emotion recognition (NLP) - Physiological signals (ECG) driving micro-expressions

5.2.3 Preference for viewing medium

The most popular option was the VR3D (approximately 62%), VR2D (approximately 22%), and mobile 2D (approximately 15%). The score difference amounted to 0.9 points, indicating a high polarization in students' acceptance of the medium. The summary of experimenters' opinions on viewing medium in VR is shown in Table 6

6 DISCUSSION AND DESIGN GUIDELINE

This study compares performance metrics of virtual reality 3D perspectives, showing the roundtable configuration outperforms fixed and row-column layouts. The roundtable achieved a mean learning preference score of 5.1, a 27.5% increase over the other two perspectives ($M=4.0$), demonstrating superior efficacy for learning outcomes. Despite the row-column design emphasizing collaborative interaction, its performance matched the fixed perspective, indicating that spatial arrangements in VR 3D environments require task-specific optimization. The roundtable's unanimous 100% preference rate among participants further validates its viability as a leading configuration for educational technology applications.

Integrating fixed and roundtable perspectives in virtual classrooms and dynamic avatars, interactivity optimizes immersion and knowledge efficiency while bridging emotion gaps.

- **Viewpoint Flexibility Enhances Learning:** Participants acknowledged the benefits of the fixed perspective for detail-oriented tasks, such as stabilizing visual focus via the "visual locking effect." For instance, Participant P32 noted its superiority for complex equations, while the round-table perspective risked obscuring details. Hybrid systems address this by incorporating explicit switching buttons, contextual triggers, and AI-driven adaptive mechanisms (e.g., locking the overhead view during teacher demonstrations or enabling multi-angle exploration during student activities) [28].
- **AI and Real-Time Adaptation Improve Engagement:** The system dynamically adjusts visual focus by prioritizing critical steps in chemistry experiments and responding to learners' attentional cues [7]. This "fusion of reality" preserves informational accuracy while leveraging spatial immersion, achieving dual optimization.
- **Avatar Emotional Realism Is Critical:** Prior research emphasizes the role of teacher interactivity in virtual classrooms. Dynamic avatars contribute to improved retention, as participants reported increased collaborative engagement during equation simulations. Conversely, low interactivity exposes technical flaws (e.g., static expressions, voice-animation mismatches), reducing empathy and learning preference. Solutions include effective computational models (e.g., micro-expression mapping [17]) and bidirectional protocols (e.g., emotion-triggered body language [18, 16]).

- **Context-Adaptive Avatars Deepen Immersion:** In history lessons, avatars using biometric cues (e.g., pupil dilation, gestures) can better convey strategic thinking than text. Developers must prioritize motion capture accuracy, real-time rendering [35], and adaptive expression frameworks (e.g., lesson-aligned micro-expression libraries [11]) for empathetic environments.

Integrating adaptive viewpoints alongside emotionally intelligent avatars culminates in a meticulously balanced and profoundly immersive educational experience, effectively bridging the gap between the precision of technical execution and the nuanced interactions characteristic of human-like engagement. In light of the empirical findings derived from the study, a comprehensive set of design guidelines tailored explicitly for the development and implementation of teaching avatars is hereby proposed for consideration. Firstly, within the context of the classroom environment where avatar technology is deployed, it is highly beneficial to significantly enhance the overall experience of students by placing a concentrated emphasis on fostering high levels of interaction and emotional engagement while simultaneously showcasing the dynamic nature of immersion that the classroom setting can provide. Secondly, concerning the various viewing perspectives available, round-table views must be actively supported to promote a heightened sense of immersion within the classroom environment, a priority that should be established in order to fully leverage the myriad advantages presented by 3D avatar-based teaching methodologies. Finally, applying a fixed viewing perspective may be utilized judiciously across all or specific segments of the lesson session, serving as a mechanism to cue essential content and effectively direct the focus of attention. Table 7 provides a synopsis of the design guidelines and supplementary commentary on our recommendations.

7 CONCLUSION

This study utilized experimental methods to investigate how interactive design, classroom perspective models, and display devices jointly influence learning experiences in virtual classroom environments. Rather than examining these factors in isolation, the study provides an integrated comparative evaluation within a single controlled experimental framework, formalized through the Virtual Classroom Interaction and Multimodal Assessment (VCIMA) framework. While several findings in this study are consistent with prior research in VR-based learning, the contribution of this work lies in its integrated comparative evaluation under a controlled experimental setup, rather than in claiming entirely novel learning effects. Within this context, the proposed framework is intended to serve as a possible analytical structure that may assist other researchers and designers in organizing and evaluating VR educational tool design.

The results demonstrate that proactive, interactive instructional models (e.g., real-time voice-based dialogue) significantly enhance

student engagement and short-term knowledge acquisition in the context of procedural calculus learning. In particular, the circular (round-table) perspective was consistently preferred, likely due to its ability to recreate spatial relationships and peer awareness within the classroom, whereas fixed perspectives required active interaction to mitigate attentional disengagement. Differences attributable solely to display devices were comparatively limited, with immersive virtual reality exhibiting the most pronounced advantages. Overall, these findings clarify how common virtual classroom design factors interact in practice within the VCIMA framework and suggest that effective virtual classroom design should prioritize dynamic interaction logic, adaptive multi-level perspectives, and emotionally lightweight avatar behaviors to balance pedagogical effectiveness with technical feasibility.

Limitations and Future Work.

Several limitations should be considered when interpreting these results. First, the instructional exposure was limited to a single 10-minute session. Accordingly, the observed learning outcomes should be interpreted as indicators of short-term comprehension and procedural performance rather than long-term retention, transfer, or sustained engagement. Future work may incorporate longitudinal designs, repeated instructional sessions, or delayed post-tests.

Second, the scope of experimental conditions was intentionally constrained to enable controlled comparison. The primary quantitative analysis focused on five core experimental scenarios. Two-dimensional avatars were evaluated only under fixed-layout conditions, and additional instructional settings (e.g., Mobile 2D and Real Classroom) were included solely as supplementary baseline references for qualitative comparison rather than as primary factors in the statistical analysis. Expanding condition coverage within a unified factorial design remains an important direction for future research.

Third, participants were university students aged 18–25 from a single cultural and educational context. While sufficient for internal validity and comparative analysis, broader demographic coverage and cross-cultural replication would be necessary to improve generalizability.

Finally, several findings in this study are consistent with prior research in VR-based learning, including the observed benefits of higher-fidelity 3D avatars and collaborative round-table classroom layouts. The contribution of this work lies in its integrated comparative evaluation under a controlled experimental setup and the VCIMA framework, rather than in claiming entirely novel outcomes. Future research may build upon this framework to explore additional pedagogical contexts, longer-term learning processes, and more diverse classroom configurations.

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